MATH 231 – Linear Algebra I

The Matrix Unit, Part 3 ▪ Matrices That Move the Plane

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About this document. This is Part 3 of the matrix unit, Sections 15–21. Section numbers run continuously through the unit, so §13.3 points into Part 2. References into the vector unit always name it.

Assumed. From Part 2: a matrix as a linear transformation, and the fact that column j of $\mathbf{A}$ is $\mathbf{A}\mathbf{e}_j$ (§13); the 2×2 determinant as an area factor with a sign (§14). From Part 1: the transpose and its laws (§6.4), symmetric matrices (§6.5), the norm written as $\mathbf{v}^\mathsf{T}\mathbf{v}$ (§7.3), and M1–M6 (§8.4). From the vector unit: length, angle, the dot product, and orthogonality (§10–§11).

What you should be able to do afterwards. Build the matrix of a scaling, a shear, a rotation by a right angle, a reflection across an axis or a diagonal, and a projection onto an axis, from where it sends $\mathbf{e}_1$ and $\mathbf{e}_2$; recognise each from the structure of its matrix; explain why no 2×2 matrix can translate, and translate with a 3×3 matrix instead; prove that a matrix preserving every length satisfies $\mathbf{A}^\mathsf{T}\mathbf{A} = \mathbf{I}$; state what an orthogonal matrix is; and show that every 2×2 orthogonal matrix is a rotation or a reflection.

15. Building a matrix from what it does

15.1. The recipe

From Part 2. A linear transformation of $\mathbb{R}^2$ is completely determined by where it sends $\mathbf{e}_1$ and $\mathbf{e}_2$, and its matrix has those two images as its columns (§13.3). That turns a geometric description into a matrix in three steps.

Step 1. Decide, from the picture, where the motion sends $\mathbf{e}_1 = [\,1, 0\,]$.

Step 2. Decide where it sends $\mathbf{e}_2 = [\,0, 1\,]$.

Step 3. Write those two vectors down as the first and second columns.

Every matrix in §16–§19 is built this way, and nothing else is needed, *provided the motion is linear*. That proviso is not automatic, so it is worth knowing how to spot a motion that fails it.

15.2. Is the motion linear?

From §13.2 and §13.4, a linear transformation of the plane

- keeps the origin where it is;
- sends lines through the origin to lines through the origin (or to the origin itself);
- sends the coordinate grid to a grid whose lines are still parallel and still evenly spaced.

A motion that breaks any of these is not linear, and no matrix performs it. Bending a line into a curve breaks the second; any motion that moves the origin breaks the first. §17 is about the most natural motion of that kind.

15.3. Reading the structure back

The recipe also runs backwards. Given a matrix, its columns say where $\mathbf{e}_1$ and $\mathbf{e}_2$ go, and the *pattern* of its entries (§4) says what kind of motion it is. The one pattern to watch from the start is the off-diagonal. In

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix},$$

the off-diagonal entries b and c are the only ones that let y influence the new first coordinate, or x the new second one. A diagonal matrix keeps the two coordinates separate. Every matrix with nonzero off-diagonal entries mixes them.

15.4. One motion after another

Two motions can be done in succession: first $\mathbf{B}$, then $\mathbf{A}$. The point $\mathbf{x}$ goes first to $\mathbf{Bx}$, then to $\mathbf{A}(\mathbf{Bx})$. By associativity (M1 of §8.4, with $\mathbf{x}$ read as an $n \times 1$ matrix),

$$\mathbf{A}(\mathbf{Bx}) \;=\; (\mathbf{AB})\mathbf{x}.$$

So *the matrix of "first* $\mathbf{B}$*, then* $\mathbf{A}$*" is the product* $\mathbf{AB}$. Mind the order: the motion done first is written on the *right*, next to the vector it acts on. Since $\mathbf{AB}$ and $\mathbf{BA}$ are usually different (§8.4), the order in which motions are done usually matters.
In the figures below, the dashed square and the grey F are the starting position. The shaded parallelogram and the blue F are where the matrix moves them. The red arrow is $\mathbf{Ae}_1$ and the green arrow is $\mathbf{Ae}_2$, which are the two columns.

16. Scaling

16.1. Uniform scaling

The motion: stretch every vector by the same factor α, so $\mathbf{x} \mapsto \alpha\mathbf{x}$.

The recipe: $\mathbf{e}_1 \mapsto \alpha\mathbf{e}_1 = [\,\alpha, 0\,]$ and $\mathbf{e}_2 \mapsto \alpha\mathbf{e}_2 = [\,0, \alpha\,]$. So the matrix is

$$\begin{bmatrix} \alpha & 0 \\ 0 & \alpha \end{bmatrix} \;=\; \alpha\mathbf{I}, \qquad \alpha\mathbf{I}\,\mathbf{x} = \alpha\mathbf{x}. \quad \checkmark$$

Every vector stays on its own line through the origin, and every length is multiplied by $|\alpha|$. With $\alpha > 1$ the plane is enlarged; with $0 < \alpha < 1$ it shrinks; with $\alpha = 0$ it collapses to the origin. With $\alpha < 0$ every vector is also turned to point the opposite way, and $\alpha = -1$ gives $-\mathbf{I}$, which §19 identifies as a rotation by 180°.

The determinant is $\det(\alpha\mathbf{I}) = \alpha \cdot \alpha - 0 = \alpha^2$. Doubling every length multiplies every area by 4, which is the familiar rule for similar figures, now read off a matrix.

16.2. Non-uniform scaling

The motion: stretch the x-direction by s_1 and the y-direction by s_2, independently: $[\,x, y\,] \mapsto [\,s_1 x,\ s_2 y\,]$.

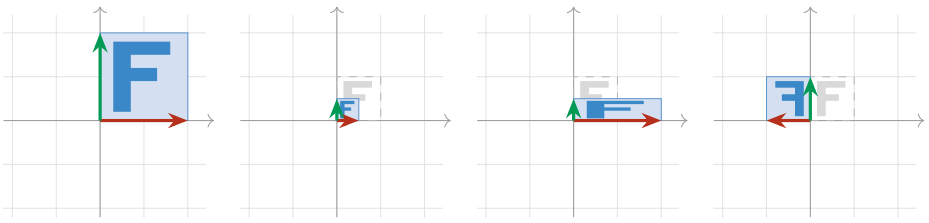
The recipe: $\mathbf{e}_1 \mapsto [\,s_1, 0\,]$ and $\mathbf{e}_2 \mapsto [\,0, s_2\,]$, so the matrix is the diagonal matrix

$$\operatorname{diag}(s_1, s_2) = \begin{bmatrix} s_1 & 0 \\ 0 & s_2 \end{bmatrix}, \qquad \det = s_1 s_2.$$

A circle becomes an ellipse. If $[\,x, y\,]$ is on the unit circle, $x^2 + y^2 = 1$, then its image $[\,u, v\,] = [\,s_1 x,\ s_2 y\,]$ satisfies

$$\frac{u^2}{s_1^2} + \frac{v^2}{s_2^2} \;=\; x^2 + y^2 \;=\; 1,$$

an ellipse with semi-axes $|s_1|$ and $|s_2|$ along the coordinate axes. A negative entry also reverses one direction: $\mathrm{diag}(-1, 1)$ sends $[\,x, y\,]$ to $[\,-x, y\,]$, the mirror image across the y-axis, and its determinant is -1.



$2\mathbf{I}$
$\det = 4$

$\frac{1}{2}\mathbf{I}$
$\det = \frac{1}{4}$

$\mathrm{diag}(2, \frac{1}{2})$
$\det = 1$

$\mathrm{diag}(-1, 1)$
$\det = -1$

The third picture is worth a second look: stretched by 2 one way and squeezed by $\frac{1}{2}$ the other, the square changes shape completely and keeps its area exactly, as $\det = 1$ says.

16.3. The structure: diagonal

A matrix is diagonal (§4.2) exactly when $\mathbf{A}\mathbf{e}_1$ is a multiple of $\mathbf{e}_1$ and $\mathbf{A}\mathbf{e}_2$ is a multiple of $\mathbf{e}_2$, that is, when each coordinate axis is sent into itself. Diagonal matrices are the scalings, and the zeros off the diagonal are the statement that the coordinates are not mixed.

17. Translation

17.1. The motion

Slide the whole plane by a fixed vector $\mathbf{t} = [\,t_1, t_2\,]$, so $\mathbf{x} \mapsto \mathbf{x} + \mathbf{t}$. Nothing turns, nothing stretches: every point moves the same distance in the same direction. It is the simplest motion there is.

17.2. No 2×2 matrix does it

Theorem 17.1. *If $\mathbf{t} \neq \mathbf{0}$, there is no matrix $\mathbf{A} \in \mathbb{R}^{2\times 2}$ with $\mathbf{A}\mathbf{x} = \mathbf{x} + \mathbf{t}$ for every $\mathbf{x} \in \mathbb{R}^2$.*

Proof. Try $\mathbf{x} = \mathbf{0}$. Every matrix sends $\mathbf{0}$ to $\mathbf{0}$, since each entry of $\mathbf{A0}$ is a dot product with the zero vector. The translation sends $\mathbf{0}$ to $\mathbf{t} \neq \mathbf{0}$. □

Translation fails the first test of §15.2: it moves the origin. It fails the definition of linearity too, since doubling the input does not double the output: $2\mathbf{x} + \mathbf{t} \neq 2(\mathbf{x} + \mathbf{t})$. Applying the recipe of §15.1 anyway would put $\mathbf{e}_1 + \mathbf{t}$ and $\mathbf{e}_2 + \mathbf{t}$ in as columns, and the resulting matrix still sends $\mathbf{0}$ to $\mathbf{0}$, so it is not the translation. That is the mistake the proviso in §15.1 guards against.

17.3. Homogeneous coordinates

There is a way around this, and it is used by every computer graphics system: add a third coordinate, always equal to 1. Represent the point $[\,x, y\,]$ of the plane by $[\,x, y, 1\,] \in \mathbb{R}^3$. Then

$$\begin{bmatrix} 1 & 0 & t_1 \\ 0 & 1 & t_2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + t_1 \\ y + t_2 \\ 1 \end{bmatrix}.$$

The first two entries are the translated point, and the third is still 1. A translation of the plane has become multiplication by a 3×3 matrix. The origin of $\mathbb{R}^3$ still stays put, as it must. What moves is the origin of the *plane*, which now lives at $[\,0, 0, 1\,]$.

Remark. Seen in $\mathbb{R}^3$, that matrix is a shear (§18): it slides each horizontal slice $z = c$ sideways by $c\,\mathbf{t}$, and the plane $z = 1$, where our copy of $\mathbb{R}^2$ lives, by exactly $\mathbf{t}$. A motion that is not linear in two dimensions becomes linear one dimension up.

17.4. Combining translation with the other motions

Any 2×2 matrix $\mathbf{M}$ fits into the same scheme as the block matrix $\begin{bmatrix} \mathbf{M} & \mathbf{0} \\ \mathbf{0}^{\mathsf{T}} & 1 \end{bmatrix}$, which applies $\mathbf{M}$ to $[\,x, y\,]$ and leaves the 1 alone. Then motions of both kinds can be chained by multiplying 3×3 matrices, in the order of §15.4.

Example 17.1. Rotate by 90° (the matrix $\mathbf{J}$ of §19.1) and *then* translate by $\mathbf{t} = [\,2, 0\,]$. The second motion is written on the left:

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 2 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

On the point $[\,2, 0\,]$: rotating gives $[\,0, 2\,]$, translating gives $[\,2, 2\,]$, and the matrix agrees: $[\,0 - 0 + 2,\ 2 + 0 + 0,\ 1\,] = [\,2, 2, 1\,]$. In the other order, translate first and then rotate, the product is $\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$. That is a different motion, as §15.4 warned.

18. Shear

18.1. The horizontal shear

The motion: hold the x-axis fixed and slide every horizontal line sideways by an amount proportional to its height. A deck of cards pushed from the side does this.

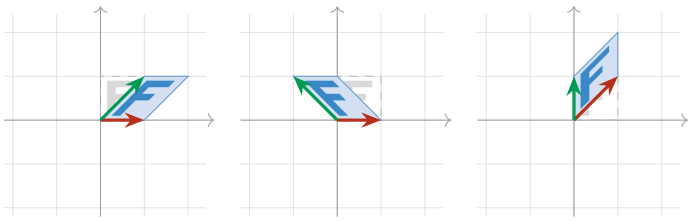
The recipe: $\mathbf{e}_1$ is on the x-axis and stays put. $\mathbf{e}_2$ is at height 1, so it slides sideways by k to $[\,k, 1\,]$. The matrix is

$$\mathbf{S} = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}, \qquad \mathbf{S} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + ky \\ y \end{bmatrix}.$$

A point at height y moves sideways by ky, as described.

18.2. The vertical shear

Holding the y-axis fixed instead and sliding vertical lines up by k times their distance across gives $\mathbf{e}_1 \mapsto [1, k]$, $\mathbf{e}_2 \mapsto \mathbf{e}_2$, and the matrix $\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$, the transpose of the horizontal shear.



$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

18.3. The structure, and the area

A shear is triangular (§4.4) with 1s on the diagonal, and $\det \mathbf{S} = 1 \cdot 1 - k \cdot 0 = 1$. The area is unchanged, and the picture says why: the square becomes a parallelogram with the same base and the same height. Lengths and angles do change. $\mathbf{e}_2$, of length 1, becomes $[k, 1]$, of length $\sqrt{1+k^2}$. A shear changes shape without changing area.

19. Rotation, reflection and projection: first examples

Three more kinds of motion, each met here in its simplest form. The rotation by a general angle is built in §24, and the projection onto a general line in §26.

19.1. Rotation by a right angle

The motion: turn the plane a quarter-turn counterclockwise about the origin.

The recipe: $\mathbf{e}_1 = [1, 0]$, pointing east, turns to point north: $[0, 1]$. $\mathbf{e}_2 = [0, 1]$, pointing north, turns to point west: $[-1, 0]$. So

$$\mathbf{J} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \qquad \mathbf{J} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}.$$

A check that this really is a quarter-turn: $[x, y]$ and $[-y, x]$ have the same length, and their dot product is $-xy + yx = 0$, so they are orthogonal.

Remark. Compare the complex numbers: multiplying $x+iy$ by i gives $-y+ix$, the same arithmetic. Multiplication by i is a quarter-turn of the complex plane, and $\mathbf{J}$ is its matrix.

A half-turn sends $\mathbf{e}_1 \mapsto -\mathbf{e}_1$ and $\mathbf{e}_2 \mapsto -\mathbf{e}_2$, which is $-\mathbf{I}$, the uniform scaling by -1 of §16.1 seen from another angle. And two quarter-turns make a half-turn: by §15.4 that is $\mathbf{JJ} = \mathbf{J}^2$, and indeed $\mathbf{J}^2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -\mathbf{I}$, the matrix version of $i^2 = -1$.

19.2. Reflections

Across the x-axis: $\mathbf{e}_1$ is on the mirror and stays; $\mathbf{e}_2$ flips to $-\mathbf{e}_2$. The matrix is $\operatorname{diag}(1, -1)$, and $[x, y] \mapsto [x, -y]$. Across the y-axis it is $\operatorname{diag}(-1, 1)$, as in §16.2.

Across the line $y = x$: the mirror makes a 45° angle with each axis, so it swaps them: $\mathbf{e}_1 \mapsto \mathbf{e}_2$ and $\mathbf{e}_2 \mapsto \mathbf{e}_1$.

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad [\,x, y\,] \mapsto [\,y, x\,].$$

The reflection across $y = x$ simply exchanges the two coordinates. Across the other diagonal, $y = -x$, the same reasoning gives $\mathbf{e}_1 \mapsto -\mathbf{e}_2$, $\mathbf{e}_2 \mapsto -\mathbf{e}_1$, and $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$.

Each reflection has determinant -1: it turns the plane over. And each one, done twice, puts everything back. For example, $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}^2 = \mathbf{I}$.

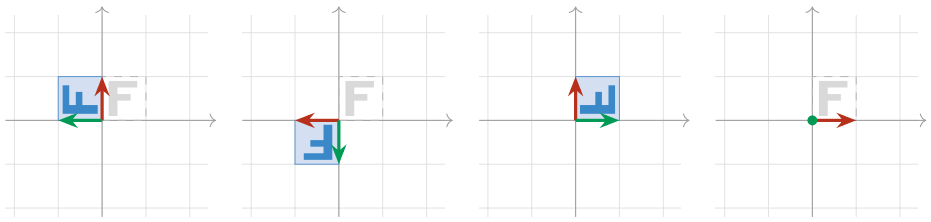
19.3. Projection onto an axis

The motion: drop every point straight down (or up) onto the x-axis: $[\,x, y\,] \mapsto [\,x, 0\,]$. Think of the shadow cast by a light directly overhead.

The recipe: $\mathbf{e}_1$ is already on the x-axis and stays; $\mathbf{e}_2$ is straight above the origin and drops to $\mathbf{0}$. So

$$\mathbf{P} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \operatorname{diag}(1, 0).$$

This is the first matrix in the gallery with determinant 0 and rank 1 (§14.2). The whole plane is flattened onto the x-axis, and the whole y-axis is crushed to the origin. Projecting a second time changes nothing, since a point already on the x-axis stays there, so $\mathbf{P}^2 = \mathbf{P}$. Part 5 takes that equation as the definition of a projection.



J
rotation by 90°

−I
rotation by 180°

reflection
across $y = x$

projection
onto the x-axis

In the last picture $\mathbf{P}\mathbf{e}_2 = \mathbf{0}$, shown as the green dot, and the F has been flattened into a segment of the x-axis.

19.4. The gallery so far

Motion	**Matrix**	det	**Structure**
uniform scaling by α	$\alpha\mathbf{I}$	α^2	multiple of $\mathbf{I}$
scaling the axes by s_1, s_2	$\mathrm{diag}(s_1, s_2)$	$s_1 s_2$	diagonal
horizontal shear	$\left[\begin{smallmatrix} 1 & k \\ 0 & 1 \end{smallmatrix}\right]$	1	triangular, unit diagonal
rotation by 90°	$\left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$	1	$\mathbf{J}^\mathsf{T} = -\mathbf{J}$
reflection across $y = x$	$\left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$	-1	symmetric, swaps coordinates
projection onto the x-axis	$\mathrm{diag}(1, 0)$	0	diagonal, rank 1
translation by $\mathbf{t} \neq \mathbf{0}$	none (2×2)	—	not linear

20. Matrices that preserve length

20.1. The question

Some motions in the gallery change lengths and some do not. A scaling by 2 doubles every length. A shear sends $\mathbf{e}_2$, of length 1, to $[\,k, 1\,]$, of length $\sqrt{1+k^2}$. A projection sends $\mathbf{e}_2$ to $\mathbf{0}$. But the quarter-turn $\mathbf{J}$ and the reflections move every vector without changing its length at all. So we ask:

> Which matrices $\mathbf{A} \in \mathbb{R}^{n \times n}$ satisfy $\|\mathbf{A}\mathbf{x}\|_2 = \|\mathbf{x}\|_2$ for every $\mathbf{x} \in \mathbb{R}^n$?

The question makes sense in any dimension, and the answer will too. Think of $n = 2$ while reading.

20.2. Rewriting the condition

Lengths are awkward to work with because of the square root, so square both sides. §7.3 wrote the squared norm as a product: $\|\mathbf{v}\|^2 = \mathbf{v}^\mathsf{T}\mathbf{v}$. Applied to $\mathbf{v} = \mathbf{A}\mathbf{x}$, with $(\mathbf{A}\mathbf{x})^\mathsf{T} = \mathbf{x}^\mathsf{T}\mathbf{A}^\mathsf{T}$ by M6,

$$\|\mathbf{A}\mathbf{x}\|^2 = (\mathbf{A}\mathbf{x})^\mathsf{T}(\mathbf{A}\mathbf{x}) = \mathbf{x}^\mathsf{T}\mathbf{A}^\mathsf{T}\mathbf{A}\mathbf{x}, \qquad \|\mathbf{x}\|^2 = \mathbf{x}^\mathsf{T}\mathbf{x} = \mathbf{x}^\mathsf{T}\mathbf{I}\mathbf{x}.$$

So the condition $\|\mathbf{A}\mathbf{x}\|^2 = \|\mathbf{x}\|^2$ for every $\mathbf{x}$ says, after subtracting and using distributivity,

$$\boxed{\mathbf{x}^\mathsf{T}\big(\mathbf{A}^\mathsf{T}\mathbf{A} - \mathbf{I}\big)\mathbf{x} = 0 \quad \text{for every } \mathbf{x} \in \mathbb{R}^n.}$$

The expression $\mathbf{x}^\mathsf{T}\mathbf{M}\mathbf{x}$ is a number (a $1 \times n$ times an $n \times n$ times an $n \times 1$). The question has become: if a matrix $\mathbf{M}$ makes this number zero for every $\mathbf{x}$, must $\mathbf{M}$ itself be zero?

20.3. The lemma

One small computation does all the work. For standard basis vectors,

$$\mathbf{e}_i^\mathsf{T}\mathbf{M}\mathbf{e}_j \;=\; m_{ij},$$

because $\mathbf{M}\mathbf{e}_j$ is column j of $\mathbf{M}$ (§13.3), and dotting with $\mathbf{e}_i$ picks out its i-th entry. So by feeding in well-chosen vectors, the number $\mathbf{x}^\mathsf{T}\mathbf{M}\mathbf{x}$ can be made to report individual entries of $\mathbf{M}$.

Lemma 20.1. *Let $\mathbf{M} \in \mathbb{R}^{n\times n}$ be symmetric, $\mathbf{M}^{\mathsf{T}} = \mathbf{M}$, and suppose $\mathbf{x}^{\mathsf{T}}\mathbf{M}\mathbf{x} = 0$ for every $\mathbf{x} \in \mathbb{R}^n$. Then $\mathbf{M} = \mathbf{0}$.*

Proof. The diagonal. Take $\mathbf{x} = \mathbf{e}_i$. Then $0 = \mathbf{e}_i^{\mathsf{T}}\mathbf{M}\mathbf{e}_i = m_{ii}$. So every diagonal entry is zero.

Off the diagonal. Take $i \neq j$ and $\mathbf{x} = \mathbf{e}_i + \mathbf{e}_j$. Expanding with the distributive laws,

$$0 = (\mathbf{e}_i + \mathbf{e}_j)^{\mathsf{T}}\mathbf{M}(\mathbf{e}_i + \mathbf{e}_j) = \mathbf{e}_i^{\mathsf{T}}\mathbf{M}\mathbf{e}_i + \mathbf{e}_i^{\mathsf{T}}\mathbf{M}\mathbf{e}_j + \mathbf{e}_j^{\mathsf{T}}\mathbf{M}\mathbf{e}_i + \mathbf{e}_j^{\mathsf{T}}\mathbf{M}\mathbf{e}_j = m_{ii} + m_{ij} + m_{ji} + m_{jj}.$$

The diagonal entries are already known to be 0, and $m_{ji} = m_{ij}$ by symmetry, so this says $2m_{ij} = 0$. Hence $m_{ij} = 0$ for every $i \neq j$ as well. □

For $n = 2$ the lemma can be seen in one line. Writing it out,

$$\mathbf{x}^{\mathsf{T}}\mathbf{M}\mathbf{x} = m_{11}x_1^2 + (m_{12} + m_{21})\,x_1x_2 + m_{22}x_2^2,$$

and the proof chose $\mathbf{x} = [\,1, 0\,]$, $[\,0, 1\,]$ and $[\,1, 1\,]$ to isolate the three coefficients one at a time.

Remark (Why symmetry is needed). The display shows that $\mathbf{x}^{\mathsf{T}}\mathbf{M}\mathbf{x}$ only ever sees the *sum* $m_{12} + m_{21}$, never the two entries separately. Without symmetry the lemma is false. The quarter-turn of §19.1 gives

$$\mathbf{x}^{\mathsf{T}}\mathbf{J}\mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix} = -x_1x_2 + x_2x_1 = 0$$

for every $\mathbf{x}$, and $\mathbf{J} \neq \mathbf{0}$. Geometrically this says $\mathbf{J}\mathbf{x} \perp \mathbf{x}$ for every $\mathbf{x}$, which is what a quarter-turn does. Matrices like $\mathbf{J}$, with $\mathbf{M}^{\mathsf{T}} = -\mathbf{M}$, are called *skew-symmetric*, and they are exactly the ones whose $\mathbf{x}^{\mathsf{T}}\mathbf{M}\mathbf{x}$ is always zero. The proof above breaks at the step $m_{ij} + m_{ji} = 0$, which for them says only $m_{ji} = -m_{ij}$.

20.4. The theorem

Theorem 20.2. *Let $\mathbf{A} \in \mathbb{R}^{n\times n}$. Then $\|\mathbf{A}\mathbf{x}\| = \|\mathbf{x}\|$ for every $\mathbf{x} \in \mathbb{R}^n$ if and only if $\mathbf{A}^{\mathsf{T}}\mathbf{A} = \mathbf{I}$.*

Proof. Suppose lengths are preserved. By §20.2 the matrix $\mathbf{M} = \mathbf{A}^{\mathsf{T}}\mathbf{A} - \mathbf{I}$ has $\mathbf{x}^{\mathsf{T}}\mathbf{M}\mathbf{x} = 0$ for every $\mathbf{x}$. It is symmetric: $(\mathbf{A}^{\mathsf{T}}\mathbf{A})^{\mathsf{T}} = \mathbf{A}^{\mathsf{T}}(\mathbf{A}^{\mathsf{T}})^{\mathsf{T}} = \mathbf{A}^{\mathsf{T}}\mathbf{A}$ by M6 and $(\mathbf{A}^{\mathsf{T}})^{\mathsf{T}} = \mathbf{A}$ (§6.4), $\mathbf{I}$ is symmetric, and the transpose of a difference is the difference of the transposes (§6.4). So the lemma applies, $\mathbf{M} = \mathbf{0}$, and $\mathbf{A}^{\mathsf{T}}\mathbf{A} = \mathbf{I}$.

Conversely, if $\mathbf{A}^{\mathsf{T}}\mathbf{A} = \mathbf{I}$ then $\|\mathbf{A}\mathbf{x}\|^2 = \mathbf{x}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \mathbf{x}^{\mathsf{T}}\mathbf{x} = \|\mathbf{x}\|^2$. □

20.5. Reading $\mathbf{A}^{\mathsf{T}}\mathbf{A} = \mathbf{I}$ column by column

The equation $\mathbf{A}^{\mathsf{T}}\mathbf{A} = \mathbf{I}$ is n^2 equations, one per entry, and each says something about the columns $\mathbf{a}_1, \ldots, \mathbf{a}_n$ of $\mathbf{A}$. Row i of $\mathbf{A}^{\mathsf{T}}$ is column i of $\mathbf{A}$, so

$$(\mathbf{A}^{\mathsf{T}}\mathbf{A})_{ij} = (\text{row } i \text{ of } \mathbf{A}^{\mathsf{T}}) \cdot (\text{column } j \text{ of } \mathbf{A}) = \mathbf{a}_i^{\mathsf{T}}\mathbf{a}_j.$$

The entries of $\mathbf{I}$ are recorded by the **Kronecker delta**,

$$\delta_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j, \end{cases}$$

so $\mathbf{A}^{\mathsf{T}}\mathbf{A} = \mathbf{I}$ says $\mathbf{a}_i^{\mathsf{T}}\mathbf{a}_j = \delta_{ij}$ for every pair i, j. Read in two halves:

- $i = j$: $\mathbf{a}_i^{\mathsf{T}}\mathbf{a}_i = \|\mathbf{a}_i\|^2 = 1$. *Every column is a unit vector.*

- $i \neq j$: $\mathbf{a}_i^{\mathsf{T}}\mathbf{a}_j = 0$. *Every two different columns are orthogonal.*

Definition 20.1 (Orthonormal)**.** Vectors $\mathbf{q}_1, \dots, \mathbf{q}_k$ are **orthonormal** if $\mathbf{q}_i^{\mathsf{T}}\mathbf{q}_j = \delta_{ij}$: each has length 1 and each two are orthogonal.

Theorem 20.3. *For $\mathbf{A} \in \mathbb{R}^{n\times n}$ the following are equivalent: $\|\mathbf{Ax}\| = \|\mathbf{x}\|$ for every $\mathbf{x}$; $\mathbf{A}^{\mathsf{T}}\mathbf{A} = \mathbf{I}$; the columns of $\mathbf{A}$ are orthonormal.*

Orthogonal nonzero vectors are independent (the vector unit, §21.2), so n orthonormal columns are an orthonormal *basis* of $\mathbb{R}^n$. And the converse direction of §20.4 shows more than lengths survive. For any $\mathbf{x}, \mathbf{y}$,

$$(\mathbf{Ax})^{\mathsf{T}}(\mathbf{Ay}) = \mathbf{x}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}\mathbf{Ay} = \mathbf{x}^{\mathsf{T}}\mathbf{y},$$

so dot products are preserved, and with them angles (the vector unit, §11.3). A length-preserving matrix moves the space *rigidly*: nothing is stretched, nothing is bent, no angle opens or closes.

20.6. Orthogonal matrices

Definition 20.2 (Orthogonal matrix)**.** A square matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ is **orthogonal** if

$$\mathbf{A}^{\mathsf{T}}\mathbf{A} = \mathbf{A}\mathbf{A}^{\mathsf{T}} = \mathbf{I}.$$

The first equation is the one §20.4 derived; it says the columns are orthonormal. The second says the *rows* are orthonormal, by the same computation applied to $\mathbf{A}^{\mathsf{T}}$. For 2×2 matrices, §21 shows that the second equation follows from the first. It does for every square matrix, though the proof needs the inverse matrix and comes in §51.3. Once that is proved, an orthogonal matrix is simply a square matrix with orthonormal columns, and the rows come free.

A common confusion ("Orthogonal" means orthonormal)*.* An orthogonal matrix has ortho*normal* columns, not merely orthogonal ones. $\mathrm{diag}(2, 1)$ has orthogonal columns, but it doubles the length of $\mathbf{e}_1$, so it is not an orthogonal matrix. The name is a historical accident, and it has stuck.

Example 20.1. From the gallery:

$$\mathbf{J}^{\mathsf{T}}\mathbf{J} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

and $\mathbf{J}\mathbf{J}^{\mathsf{T}} = \mathbf{I}$ likewise, so $\mathbf{J}$ is orthogonal. So are $-\mathbf{I}$, $\mathrm{diag}(1, -1)$, and $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$: each has columns that are unit vectors at right angles. The scaling $2\mathbf{I}$, the shear and the projection $\mathrm{diag}(1, 0)$ are not.

21. What an orthogonal matrix can do

In the plane, the condition "orthonormal columns" is restrictive enough to list every possibility.

21.1. The second column is nearly determined by the first

Lemma 21.1. *Let $[\,a, c\,] \neq \mathbf{0}$. The vectors of $\mathbb{R}^2$ orthogonal to $[\,a, c\,]$ are exactly the multiples of $[\,-c, a\,]$.*

Proof. First, $[\,-c, a\,] \cdot [\,a, c\,] = -ca + ac = 0$, so every multiple is orthogonal. Conversely, $[\,a, c\,]$ and $[\,-c, a\,]$ are nonzero and orthogonal, hence independent (the vector unit, §21.2), hence a basis of $\mathbb{R}^2$ (§18.5). So a vector $\mathbf{w}$ orthogonal to $[\,a, c\,]$ can be written $\mathbf{w} = s[\,a, c\,] + t[\,-c, a\,]$. Dotting with $[\,a, c\,]$ gives $0 = s(a^2 + c^2) + 0$, and $a^2 + c^2 \neq 0$, so $s = 0$ and $\mathbf{w} = t[\,-c, a\,]$. □

Notice that $[\,-c, a\,] = \mathbf{J}[\,a, c\,]$: the orthogonal direction is the first vector turned a quarter-turn.

21.2. Every 2×2 orthogonal matrix

Let $\mathbf{A}$ be 2×2 with $\mathbf{A}^\mathsf{T}\mathbf{A} = \mathbf{I}$. Its first column $[\,a, c\,]$ is a unit vector, so $a^2+c^2 = 1$. Its second column is a unit vector orthogonal to the first, so by the lemma it is $t[\,-c, a\,]$ with $|t| \cdot \|[\,-c, a\,]\| = |t| = 1$. That leaves exactly two choices, $t = 1$ and $t = -1$.

Theorem 21.2. *Every $\mathbf{A} \in \mathbb{R}^{2\times2}$ with $\mathbf{A}^\mathsf{T}\mathbf{A} = \mathbf{I}$ has one of the two forms*

$$\begin{bmatrix} a & -c \\ c & a \end{bmatrix} \quad (\det = a^2 + c^2 = 1) \qquad \textit{or} \qquad \begin{bmatrix} a & c \\ c & -a \end{bmatrix} \quad (\det = -a^2 - c^2 = -1),$$

with $a^2 + c^2 = 1$. Both forms also satisfy $\mathbf{A}\mathbf{A}^\mathsf{T} = \mathbf{I}$, so every such $\mathbf{A}$ is orthogonal.

Proof. The two forms were derived above. For the last claim, multiply out:

$$\begin{bmatrix} a & -c \\ c & a \end{bmatrix}\begin{bmatrix} a & c \\ -c & a \end{bmatrix} = \begin{bmatrix} a^2+c^2 & ac-ca \\ ca-ac & c^2+a^2 \end{bmatrix} = \mathbf{I}, \qquad \begin{bmatrix} a & c \\ c & -a \end{bmatrix}\begin{bmatrix} a & c \\ c & -a \end{bmatrix} = \begin{bmatrix} a^2+c^2 & ac-ca \\ ca-ac & c^2+a^2 \end{bmatrix} = \mathbf{I}.$$

(The second matrix is symmetric, so it is its own transpose.) □

So in the plane, $\mathbf{A}^\mathsf{T}\mathbf{A} = \mathbf{I}$ alone forces $\mathbf{A}\mathbf{A}^\mathsf{T} = \mathbf{I}$, which is the promise of §20.6 kept for $n = 2$.

21.3. The two families

Determinant $+1$: rotations. In $\begin{bmatrix} a & -c \\ c & a \end{bmatrix}$, the first column $[\,a, c\,]$ is any point of the unit circle, and the second is that point turned a further quarter-turn. The pair $\mathbf{e}_1, \mathbf{e}_2$ is turned rigidly to the pair $[\,a, c\,], [\,-c, a\,]$, with orientation kept. The family contains $\mathbf{I}$ ($a = 1, c = 0$), $\mathbf{J}$ ($a = 0, c = 1$), and $-\mathbf{I}$ ($a = -1, c = 0$). These are the rotations, and §24 writes them in terms of the angle of rotation.

Determinant -1: reflections. The family $\begin{bmatrix} a & c \\ c & -a \end{bmatrix}$ contains $\mathrm{diag}(1, -1)$ ($a = 1, c = 0$), $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ ($a = 0, c = 1$) and $\mathrm{diag}(-1, 1)$ ($a = -1, c = 0$), the three reflections of the gallery. Every member is a reflection across some line through the origin, and the line can be found. Using $a^2 + c^2 = 1$,

$$\begin{bmatrix} a & c \\ c & -a \end{bmatrix}\begin{bmatrix} 1+a \\ c \end{bmatrix} = \begin{bmatrix} a+a^2+c^2 \\ c+ac-ac \end{bmatrix} = \begin{bmatrix} 1+a \\ c \end{bmatrix},$$

so the vector $[\,1 + a,\, c\,]$ is left exactly where it is. That vector lies on the mirror. For $\mathrm{diag}(1, -1)$ it is $[\,2, 0\,]$, on the x-axis. For $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ it is $[\,1, 1\,]$, on the line $y = x$. (When $a = -1$ that vector is $\mathbf{0}$, and the mirror is the y-axis.)

Each reflection undone by itself: squaring $\begin{bmatrix} a & c \\ c & -a \end{bmatrix}$ gives $\mathbf{I}$, as computed in the proof above.

21.4. The gallery, sorted

Matrix	Orthogonal?	det	Family
$\mathbf{I}$, $\mathbf{J}$, $-\mathbf{I}$	yes	$+1$	rotations
$\mathrm{diag}(1,-1)$, $\mathrm{diag}(-1,1)$, $\left[\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right]$	yes	-1	reflections
$\alpha\mathbf{I}$ with $\lvert\alpha\rvert \neq 1$	no	α^2	changes every length
shear $\left[\begin{smallmatrix}1&k\\0&1\end{smallmatrix}\right]$, $k \neq 0$	no	1	changes the length of $\mathbf{e}_2$
$\mathrm{diag}(1,0)$	no	0	crushes $\mathbf{e}_2$ to $\mathbf{0}$

The shear is a useful warning: its determinant is 1, the same as a rotation's, and it is not orthogonal. A determinant of ± 1 means area is preserved, which is much weaker than every length being preserved.

21.5. Practice

P1. Find the matrix that sends $\mathbf{e}_1$ to $[\,0, 2\,]$ and $\mathbf{e}_2$ to $[\,-1, 0\,]$. Show that it is a quarter-turn followed by a scaling of the y-direction, by writing it as a product.

P2. Is $T([\,x, y\,]) = [\,x + y,\ 1\,]$ linear? Justify.

P3. Write the 3×3 matrix, in homogeneous coordinates, that translates by $[\,-1, 3\,]$, and check it on the point $[\,2, 2\,]$.

P4. Show that the vertical shear $\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$ has determinant 1, and find the image of the unit square.

P5. Which of these are orthogonal? For each 2×2 one that is, say whether it is a rotation or a reflection, and for a reflection find a vector on the mirror.

$$\begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix}, \qquad \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & -\frac{3}{5} \end{bmatrix}, \qquad \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \qquad \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

P6. Let $\mathbf{K} = \begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix}$. Show that $\mathbf{x}^{\mathsf{T}}\mathbf{K}\mathbf{x} = 0$ for every $\mathbf{x}$, and explain why this does not contradict Lemma 20.1.

Answers

P1. The columns are the images: $\begin{bmatrix} 0 & -1 \\ 2 & 0 \end{bmatrix}$. A quarter-turn first, then $\mathrm{diag}(1, 2)$: by §15.4 the product is $\mathrm{diag}(1,2)\,\mathbf{J} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 2 & 0 \end{bmatrix}$. ✓

P2. No: $T(\mathbf{0}) = [\,0, 1\,] \neq \mathbf{0}$, and a linear transformation fixes the origin (§13.2).

P3. $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$, and $[\,2, 2, 1\,] \mapsto [\,2 - 1,\ 2 + 3,\ 1\,] = [\,1, 5, 1\,]$, which represents $[\,1, 5\,] = [\,2, 2\,] + [\,-1, 3\,]$. ✓

P4. $\det = 1 \cdot 1 - 0 \cdot k = 1$. The corners $\mathbf{0}$, $\mathbf{e}_1$, $\mathbf{e}_1 + \mathbf{e}_2$, $\mathbf{e}_2$ go to $\mathbf{0}$, $[1, k]$, $[1, k+1]$, $[0, 1]$: a parallelogram with a vertical side of length 1 on the y-axis and the opposite side raised by k, area 1.

P5. The first: columns are unit vectors ($\frac{9}{25} + \frac{16}{25} = 1$) and orthogonal ($-\frac{12}{25} + \frac{12}{25} = 0$), of the form $\begin{bmatrix} a & -c \\ c & a \end{bmatrix}$ with $\det = 1$: a rotation. The second: orthogonal, of the form $\begin{bmatrix} a & c \\ c & -a \end{bmatrix}$ with $\det = -1$: a reflection, with $[1 + a, c] = [\frac{8}{5}, \frac{4}{5}]$, i.e. the direction $[2, 1]$, on the mirror. The third: its columns are orthogonal but have length $\sqrt{2}$, so it is not orthogonal (it is $\sqrt{2}$ times a rotation). The fourth: its columns are $\mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_1$, which are orthonormal, so it is orthogonal.

P6. $\mathbf{x}^\mathsf{T}\mathbf{K}\mathbf{x} = x_1(3x_2) + x_2(-3x_1) = 0$. The lemma assumes $\mathbf{M}$ is symmetric, and $\mathbf{K}$ is not: $\mathbf{K}^\mathsf{T} = -\mathbf{K}$, so $\mathbf{K}$ is skew-symmetric. $\mathbf{K} = -3\mathbf{J}$, and $\mathbf{K}\mathbf{x}$ is always orthogonal to $\mathbf{x}$.

Looking ahead. The determinants in this part were computed one matrix at a time. Part 4 collects their general properties: what happens to det under scaling, transposing, and multiplying. It then proves the claim of §14.3 that $|\det \mathbf{A}|$ is an area, from base times height. With that, it builds the rotation by an arbitrary angle and identifies it with the first family of §21.3.